

Storm clouds

Dense, dark clouds gathering overhead mean stormy weather is on the way. Thunderstorms have terrifying power but also an awesome beauty.

Our weather is created by changes in the atmosphere. When air turns cold, it sinks, becoming compressed under its own weight and causing high pressure at Earth's surface. As the air molecules squeeze together, they heat up. The warm air rises, surface pressure drops, and fair weather may follow. But when rapidly rising warm air meets descending cold air, the atmosphere becomes unsettled. Water vapor in the air turns into clouds, the clouds collide, and electric energy builds up. The electricity is released in lightning bolts that strike Earth's surface with cracks of thunder, often accompanied by heavy rainfall.

Supercell storms

One of the most dangerous weather conditions is the supercell storm, when a huge mass of cloud develops a rotating updraft of air, called a mesocyclone, at its center. The cloud cover may stretch from horizon to horizon. Above this, unseen from the ground, a cloud formation known as cumulonimbus towers like a monstrous, flat-topped mushroom into the upper atmosphere. A supercell storm system can rage for many hours, producing destructive winds, torrential rain, and giant hailstones.

